



DUST CLOUDS IN CARINA

Fantasy-like structures are found throughout the Carina Nebula, a vast molecular cloud of star birth that spans across 300 light-years of space. This detailed color-enhanced view shows a small part, roughly 15 light-years wide. The fantastic shapes are sculpted out of the cold cloud by stellar winds and ultra-violet radiation emitted by massive stars, as they slowly erode it away. Dark knots of gas and dust are so thick they are opaque, although the cloud is typically

less dense than Earth's atmosphere. The dark pillar of cold hydrogen and dust to the right of this image is more than two light-years long, and has so far resisted being worn away. Inside it new stars are taking shape. The image was taken by the Hubble Space Telescope and combines two sets of observations. The first, from 2005, were taken in light emitted by hydrogen atoms. The second, taken in 2010, were in light emitted by oxygen atoms.